Transferable Exposure Escrow Under Ambiguous Agent Effects

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Abstract

When a disconnected agent receives no receipt, its effect or rights transfer may already have succeeded. A single conservation invariant keeps both operations backed until their outcomes are durable. After rollback, the largest compatible old exposure determines how much quota can be issued again. A matching certificate sharpens the reserve when unresolved effects compete for two exclusive rights. We check the invariant by finite state exploration and exercise lost acknowledgments in two local SQLite stores. In the database control, a timeout refund exceeds the exposure cap while the quota ledger still balances. Applying the rule requires durable custody and enforcement at the actual effect sink.

Summary and scope

Question. How can an agent enforce a physical exposure cap while effects, transfers and recovery acknowledgments are delayed?

Finding. Keep each unresolved effect or transfer backed until its durable outcome is known. After closing the old generation, reserve the largest old exposure consistent with the evidence and reissue only the difference from the cap.

Evidence and scope. The invariants are proved and checked in separate finite interleaving models. A deployed system still needs enforcement at the actual sinks and a recovery register outside the state that can roll back.

1 An exposure cap that survives a network failure

A service may allow a local agent to send a bounded number of human-facing notifications while disconnected. If an acknowledgment does not return, the local service cannot infer that nothing happened. The same uncertainty applies if the service moved some of its allowance to another device. Treating either timeout as unused capacity makes a shared limit unreliable.

Escrow allocation permits local decisions under numeric constraints [1]. Bounded-counter replication applies the same principle to rights transfer between replicas [2]. Here the spend being limited is a physical effect at a separate sink. We distinguish the capacity that a ledger represents from the exposure that a person or external system has actually received.

Let B be a fixed cap. Allocate free rights among shards, totaling B. All charges are nonnegative. An effect request has a unique scoped identifier and an upper charge r that covers all effects possible under that request. Its true charge a is at most r. A transfer ticket similarly has a unique identifier, an amount, a source, a destination, and authenticated metadata. IDs bind all effect or ticket parameters, including the intended recipient. Each shard and sink has durable state that cannot silently revert or be cloned into a second spendable copy.

2 Two protocols with one accounting rule

Theorem 1. *Let F be the total free rights, R the total pending effect reservations, C the total settled charges, and T the total amounts in transit. Under the stated protocol,*

$$F + R + C + T = B.$$

If X is actual cumulative physical exposure, then

$$X \leq R + C \leq B.$$

The two protocols below preserve this accounting. The proof follows after the transitions are specified.

An effect admission atomically moves r from the owning shard's free rights into a pending reservation. A sink must enforce that reservation and atomically either commit the authorized effect once or fence the request against future execution. A successful commit binds the actual charge and result to the request. A fence attests that no effect committed and none can commit later under the same request. The shard may settle the corresponding durable outcome at most once. Settlement removes the pending amount, records a as settled, and returns $r - a$ to free rights. A fence settles with charge zero.

Transfer creation atomically debits the sender's free rights and places the amount in transit. The receiving ledger may accept the ticket once and move the amount to its free rights. Alternatively it can durably reject the ticket, ruling out acceptance. Only a matching rejection permits the sender to reclaim the amount, at most once. Acceptance and rejection are mutually exclusive. A duplicate message can return the established status. It cannot change rights. A ledger's local view may lag the recipient's decision. For the invariant, a transfer remains in transit exactly until acceptance or reclaim. An accepted transfer whose acknowledgment is missing does not make the debited rights available to the sender.

Proof. Admission moves rights from F to R. A sink commit changes physical exposure without moving the ledger reservation. A fence also leaves the reservation backed until settlement. Settlement moves r out of R, moves a into C, and restores $r - a$ to F. A ticket debit moves rights from F to T. Acceptance or reclaim moves the ticket amount back to F at the appropriate shard. A rejection alone moves no rights. Each transition preserves the sum and all four terms stay nonnegative.

Every committed effect has either its full upper charge in R or its exact charge in C. The categories are mutually exclusive by settlement and identifier uniqueness. A reservation may back an uncommitted request, but it can back no more than one logical effect. Adding the per-effect inequalities gives the physical bound. □

The atomicity requirement is at the effect boundary. Recording a key before a separate, fallible external effect can block a valid retry. Recording it afterward can permit a duplicate. Transaction commit addresses related agreement problems [3]. If an external sink lacks an enforceable commit or fence, the shard cannot use this protocol to release an ambiguous reservation.

3 Why elapsed time cannot establish an unused right

Proposition 1. *Under an asynchronous channel with no deadline enforced at the recipient, a sender cannot infer unused transfer rights from its local timeout alone. The same holds for an unresolved effect reservation at a separate sink.*

Proof. Compare two executions in which the sender transmits the same ticket and receives no response before timeout. In one execution nothing has arrived at the recipient. In the other, the recipient has accepted the ticket and its acknowledgment is delayed. The sender has the same observations. Reclaiming in the second execution permits both ledgers to spend the same rights. Even the first execution is unsafe if the ticket can arrive later and be accepted. For an effect, compare the corresponding worlds of an undelivered request and a committed request with a lost acknowledgment. Returning the pending amount is unsafe when the previous request has committed or can still commit. □

A lease can rule out future execution when expiry is enforced at the sink. Expiry alone cannot establish whether an earlier effect committed. If neither a confirmed result nor a fence is available, the protocol may strand rights for the length of the partition. It preserves the cap while leaving that availability question open.

4 Reissuing quota after snapshot rollback

A restored snapshot may show free rights that were spent after it was taken. State-continuity systems such as Memoir address the related problem of detecting rollback of protected state [4]. Here the snapshot can help identify requests, but cannot authorize new spending. First close the old generation at all effect sinks and rights recipients. Closure must fence old tokens and transfers in the same durable order as acceptance, so old physical exposure X_{old} is fixed before reissue.

Let $\mathcal{X}$ be the nonempty set of old exposures consistent with authenticated terminal observations and a complete authorization inventory. Let $U = \sup \mathcal{X}$, with $0 \le U \le B$. When request reservations are disjoint, an exact receipt contributes its charge, an authenticated final absence contributes zero, and an unresolved request contributes its reservation ceiling. Summing gives a valid upper bound. It equals U when the unresolved charges can attain their ceilings together. If the inventory is incomplete, include an upper bound for the missing authority. The restored snapshot's free balance does not supply such a bound.

Proposition 2. *The maximum quota that can be reissued and spent independently of the remaining uncertainty is $B - U$. Let G be the total quota reissued so far. As consistent terminal evidence lowers U, atomically increase G to $B - U$ and issue only the increment. The accounting record and the issuance must survive rollback together.*

Proof. If $G \le B - U$, then old exposure is at most U and new exposure is at most G by the new generation's escrow. Their sum is at most B. For any grant larger than $B - U$, the definition of supremum gives a compatible old exposure that would exceed the cap if the grant were fully spent. This proves maximality. Compatible evidence can only reduce $\mathcal{X}$, so $B - U$ can only increase. Issuing the difference from the durable total preserves the inequality, and replay of the same or a weaker observation issues no additional rights. □

If the grant is issued only in quanta of size $\eta > 0$, the largest issuable total is $\eta\lfloor (B-U)/\eta \rfloor$. The register uses that rounded total before calculating the increment. Terminal evidence must describe the same closed generation, cap and inventory. Combine valid observations by intersection. An empty compatible set signals an evidence failure and cannot authorize further issuance.

For a cap of three and reservations of two and one, suppose the only verified terminal record says the one-unit request did not commit. The two-unit request may have committed in full, so recovery can safely reissue one unit. It can issue another only after terminal evidence reduces

the bound on the two-unit request. If restoring the reissue register makes it forget the first unit, minting that unit again can violate the cap even though the old generation stays fenced.

This reissued unit funds an authorized fresh effect. It does not establish that the two-unit request was absent and cannot justify retrying that request under a new identifier. Reissue preserves the aggregate budget. A uniqueness requirement still needs terminal occupancy evidence for the particular effect.

The recovery model explored 67 reachable states and independently enumerated compatible worlds for all 12 partial terminal reports. The proposed mechanism stayed within the physical cap. Four mutants exceeded it: treating unknown as unused, using absence observed before the fence, issuing the full recoverable amount twice, and rolling back the issuance register. Each witness uses physical exposure four under cap three. This check models the durability condition. It does not demonstrate an implementation of a rollback-resistant register.

5 Reissue when unresolved requests compete

The ceilings of unresolved requests can substantially overstate what could have committed together. Suppose each request uses one exclusive grant key and one exclusive effect key. A trusted old-generation sink could commit at most once at each key. An authenticated final absence removes the associated request. A verified commit contributes its actual charge to K and occupies both keys, even if its charge is zero. Remove those occupied keys from the unresolved inventory. This step requires durable evidence that the exclusivity rules held before closure. Two verified commits on one exclusive key contradict that model.

Use a bipartite graph with grant keys on one side and effect keys on the other. An unresolved request is an edge with its charge ceiling r_e. Any jointly committed set forms a matching. Let M be the weight of a maximum matching. Then $\min(B, K + M)$ bounds the old exposure. If the constraints of the residual evidence are exactly the matching constraints and the matching ceilings are attainable together under the old cap, $K + M$ is the sharp old maximum.

A checker can certify the matching optimum without running the solver. Supply a matching and nonnegative prices α_u and β_v on the two key families. Check $\alpha_u + \beta_v \geq r_{uv}$ for every unresolved edge. Every matching uses each vertex at most once, so its total weight is at most the sum of all prices. Equality with the supplied matching proves optimality. Such a certificate exists by linear-programming duality. The primal maximizes weighted edge usage under unit capacity at each vertex. Negate the rows of its incidence matrix on one side to obtain one positive and one negative entry per column. Every square submatrix has determinant zero or ± 1, by expanding a column with at most one entry or, if every column has two, summing its rows. The unit-capacity polytope therefore has integral vertices. The dual variables are precisely the vertex prices.

For example, take a cap of 12 and one confirmed unit of exposure. In the residual inventory, the first grant key has edges of weights five and four to two effect keys. The second grant key has one edge of weight four to the first effect key. The two four-unit requests can commit together, and grant prices $(4, 3)$ and effect prices $(1, 0)$ certify the maximum of eight. The old exposure can reach nine and three units can be reissued. Summing all three unresolved ceilings would leave no quota to reissue after clipping to the old cap.

The graph gives a safe upper bound even when additional constraints make some matchings unattainable. For example, two disjoint requests each with indivisible charge four and an old cap of five can have old exposure at most four. Clipping the unconstrained matching weight to five is safe and conservative. The matching verifier agrees with independent vertex-price search on 384 synthetic instances. It does not establish that a real authorization inventory is complete.

6 A complete finite interleaving check

The verifier models two shards with total cap three, initial free rights two and one, four scoped effect requests, and a one-unit ticket in each direction. It explores all reachable states and records ledger liability separately from physically committed effects. Redeliveries that return existing status add no state transition. The model includes delayed settlement and allows a fenced effect to return its full reservation.

Protocol	States	Transitions	Maximum physical exposure
Backed until durable outcome	2,592	6,648	3
Effect timeout refunds rights	12,192	37,936	5
Transfer timeout reclaims rights	8,436	24,256	5
Recipient accepts a ticket twice	6,240	19,184	5

The safe graph has neither a conservation error nor an unbacked physical effect. Every mutant reaches a state with physical exposure four or higher. One short transfer trace sends a one-unit ticket, refunds it at timeout, and uses the sender's restored balance to commit two units. The recipient then accepts the delayed ticket and commits two more units. With cap three, that yields a physical violation even though each shard observes enough free rights locally.

7 Retrying effects across separate durable stores

The local experiment used one SQLite database for the escrow ledger and another for effect outcomes. An effect is a committed row at the sink. The ledger reserves the request ceiling in one write transaction. At the sink, the request key and parameters identify one terminal outcome, either a committed row or a fence. Replaying the same request returns the recorded outcome. Reusing the key with a different ceiling is rejected. Settlement matches the receipt to the reservation and credits the difference between its ceiling and charge in one write transaction.

The cap was five. A request reserved four and committed an effect of charge three. A child process exited without returning the receipt. Reopening both stores showed one free unit, four pending, and three committed at the sink. A two-unit request still failed admission. Retrying returned the existing receipt, which released one unit after settlement. A second request then committed two units, bringing exposure to five. Exiting after the first settlement commit and retrying did not release the unit twice.

A separate exit before the sink commit left no effect. A durable fence prevented a delayed retry from creating one. Exiting before settling that fence left the reservation in place, and a retry returned the full ceiling once. In the negative control, the ledger treated the missing reply after a successful commit as an absent effect. That allowed it to spend five additional units. The ledger still balanced, while the sink held eight units against the cap of five.

The fixture used SQLite 3.54.0 with write-ahead logging and five forced process exits. It measures transactions on disposable local rows. Real delivery requires binding the same effect identifier and terminal record at the service that performs the action. This experiment supplies neither that external transaction boundary nor evidence about hardware power loss.

8 Application and evidence boundary

The protocol can govern budgets for outbound notifications, sandboxed changes, or other effects with a meaningful upper charge. Choose the charge unit before admitting requests and log both

potential liability and committed outcomes. A prospective exercise should delay acknowledgments, interrupt settlement, replay transfers, and crash the sink at the effect boundary. Measure stranded capacity and safe throughput as well as cap violations. A ledger rollback, false upper charge, or effect that bypasses the enforcement point invalidates the bound. This interleaving model and the local transaction experiment establish different parts of the protocol. A deployed network still needs an enforced upper charge and matching receipts from the real sink.

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